

UNIT 4

BIG DATA

No.	Main Topic	Subtopics
1	Introduction to Big Data	Definition, Characteristics, Need for Big Data, Traditional Data vs Big Data
2	Characteristics of Big Data	5Vs – Volume, Velocity, Variety, Veracity, Value
3	Big Data Types	Structured Data, Semi-Structured Data, Unstructured Data
4	Distributed File System	Need for Distributed Storage, Data Blocks, Data Distribution, Replication, Fault Tolerance, Scalability
5	Big Data Architecture	Data Sources, Data Ingestion, Storage, Processing, Analytics, Visualization
6	Drivers for Big Data	Internet, Social Media, Cloud Computing, Mobile Devices, IoT, Sensors, Digital Transactions
7	Big Data Analytics	Data Collection, Data Processing, Data Analysis, Data Visualization, Decision Making
8	Types of Big Data Analytics	Descriptive, Diagnostic, Predictive, Prescriptive Analytics
9	Data Appliances	Definition, Components, Features, Advantages, Applications
10	Data Integration Tools	Data Integration, ETL, Extract, Transform, Load, Data Cleaning, Data Consolidation
11	Big Data Security	Security Challenges, Authentication, Authorization, Encryption, Access Control, Data Privacy
12	Security Principles	Confidentiality, Integrity, Availability (CIA Triad)
13	Issues in Big Data	Storage, Processing, Scalability, Data Quality, Data Integration, Security, Privacy, Real-Time Processing, Cost
14	Big Data Applications	Healthcare, Banking & Finance, Retail, Transportation, Social Media, Smart Cities, Education, Manufacturing

Big Data

Big Data = Data that is too large, too fast, too diverse, or too complex for traditional data-processing systems to handle efficiently.

Traditional Data	Big Data
Relatively smaller	Extremely large
Mostly structured	Structured + semi-structured + unstructured

Usually centralized	Often distributed
Traditional databases	Distributed storage systems
Limited processing requirements	Large-scale/parallel processing

example

YouTube.

Every day:

- Millions of videos
- Millions of comments
- Likes/dislikes
- Search history
- Watch history
- User profiles
- Advertisements

All these together create Big Data.

Big Data is not simply “a large amount of data.”

It is data with characteristics such as **Volume, Velocity, Variety, Veracity and Value.**

Volume

Volume = How much data?

It refers to the **quantity of data.**

example

A social media platform stores:

Photos	→ TBs
Videos	→ PBs
Messages	→ TBs
User data	→ TBs

If the amount of data is extremely large → **Volume**

Exam clue

If the question says:

"The organization has 500 TB of customer data."

VOLUME

Velocity

Velocity = How fast is data generated or processed?

example

Stock market:

Buy/Sell transactions
↓
Thousands/millions
of transactions
per second

The important issue is not just the amount of data but **the speed at which it arrives.**

Exam clue

Words such as:

- Real-time
- Continuous
- Transactions/second
- High-speed data generation
- Streaming

→ **VELOCITY**

Variety

Variety = Different types/forms of data.

Real-life example — Hospital

A hospital may have:

Patient table	→ Structured
Doctor's notes	→ Text
MRI scan	→ Image
ECG	→ Signal
Patient recording	→ Audio
Medical video	→ Video

Different forms of data → **VARIETY**

Exam clue

If the question says:

"Text, images, audio, video and sensor data are processed together."

Answer:

VARIETY

Veracity

Veracity = How trustworthy, accurate and reliable is the data?

example

Suppose a company collects social-media information.

Some data may be:

Correct information
Fake accounts
Spam
Duplicate records
Missing values
Incorrect information

Exam clue

Words such as:

- Accuracy
- Reliability
- Trust
- Noise
- Fake data
- Missing data
- Uncertainty

→ **VERACITY**

Value

Value = What useful benefit can we obtain from the data?

Having 100 TB of data is useless if we cannot extract anything useful from it.

example — Amazon

Amazon analyzes:

```
Search history
      +
Purchase history
      +
Product reviews
      +
Browsing behavior
      ↓
Recommendation
```

The useful outcome is **Value**.

Exam clue

If the question asks:

"How can an organization obtain useful business insights from data?"

VALUE

Structured Data

What is it?

Data that follows a **fixed structure**, usually rows and columns.

example

College database:

Roll No	Name	Department	Mark
101	Arun	CSE	85
102	Priya	IT	91

This can easily be stored in a relational database.

→ **STRUCTURED DATA**

Table = Structured

Unstructured Data

Data that doesn't have a predefined tabular structure.

Real-life examples

- YouTube video
- MRI image
- Voice recording
- Photograph
- PDF
- Social-media post
- CCTV video

Image / Video / Audio = Usually unstructured

Semi-Structured Data

Data that does not follow a strict table structure but contains **tags, keys, or metadata** that provide organization.

Examples

JSON

```
{
  "name": "Arun",
  "age": 21,
  "department": "CSE"
}
```

XML

```
<student>
  <name>Arun</name>
  <age>21</age>
</student>
```

Type	Example
Structured	SQL table
Semi-structured	JSON, XML
Unstructured	Image, video, audio

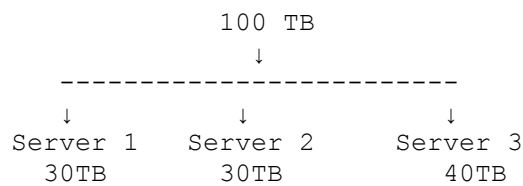
Distributed File System

A **Distributed File System (DFS)** stores data across **multiple computers connected through a network**.

Suppose you have:

100 TB of data

A single computer may not be able to efficiently store and process it.



Data is distributed across multiple machines.

example

HDFS — Hadoop Distributed File System

HDFS is designed to store very large datasets across clusters of machines.

Why Distributed Storage?

Suppose:

```
One computer
  ↓
  X
Computer fails
  ↓
Data unavailable
```

Instead:

```
Block A → Machine 1
        → Machine 2
        → Machine 3
```

If Machine 1 fails:

Machine 2/3 still have copies.

This is called:

Replication

Replication = Keeping multiple copies of data.

Fault Tolerance

The ability of a system to **continue functioning even when some components fail.**

Real-life example

Suppose a student's file is stored on:

Server A
Server B
Server C

Server A fails.

The file can still be accessed from B or C.

→ **Fault tolerance**

Failure happens → System continues working = Fault tolerance

Scalability

The ability of a system to handle increasing workload/data.

example

Today:

1 TB data
→ 10 servers

Next year:

100 TB data
→ Add more servers

The system grows with the workload.

→ **Scalability**

Horizontal Scaling

Adding **more machines**.

Before:

Server 1

After:

Server 1 + Server 2 + Server 3 + Server 4

→ Horizontal scaling

Real-life example

A Big Data cluster adds more nodes as data grows.

Vertical Scaling

Increasing the resources of an existing machine.

Before:

Server

8 GB RAM

4 CPU

After:

Server

64 GB RAM

16 CPU

→ Vertical scaling

Horizontal = More machines

Vertical = Bigger machine

Big Data Analytics

Using techniques to analyze large datasets and extract **useful information, patterns and insights**.

example

A bank analyzes millions of transactions:

Transactions
↓
Analytics
↓
Unusual transaction
↓
Possible fraud
↓
Alert

Descriptive Analytics

WHAT happened?

Example

A company asks:

"How many products did we sell last month?"

Answer:

1,25,000 products

→ Descriptive analytics

Diagnostic Analytics

WHY did it happen?

Example

Sales dropped by 20%.

The company investigates:

Why?

Maybe:

- Price increased
- Competitor launched a product
- Supply problem

→ Diagnostic analytics

Predictive Analytics

WHAT is likely to happen?

Example

A hospital analyzes historical patient data.

It predicts:

"Patient admissions are likely to increase next month."

→ Predictive analytics

Prescriptive Analytics

WHAT SHOULD WE DO?

Example

A hospital predicts increased admissions.

The system recommends:

"Increase staff and allocate additional beds."

→ Prescriptive analytics

ETL

ETL = Extract → Transform → Load

A process for bringing data from different sources into a target system.

example

Suppose a company has:

Sales Database

+
Website
+
Mobile App
+
Customer Database

Extract

Collect data from these sources.

↓

Transform

Clean/convert the data.

For example:

"Chennai"
"CHENNAI"
"chennai"

Convert them into a consistent format.

↓

Load

Put the processed data into the target system.

Sources
↓
Extract
↓
Transform
↓
Load
↓
Data Warehouse / Big Data Platform

Data Integration

Combining data from **different sources**.

Example

A university has:

Admission system
+

Attendance system
+
Exam system
+
Fee system
↓
Integrated Student Data

That's **Data Integration**.

Difference

Data Integration = Combining data

ETL = One process used to extract, transform and load data

Data Appliance

A **preconfigured hardware + software system** designed for a particular data workload.

Instead of buying and configuring everything separately:

Hardware
+
Storage
+
Database/software
+
Networking
+
Management

comes as an integrated solution.

Simple real-life analogy

Instead of buying:

CPU + motherboard + RAM + storage + OS separately

you buy a **ready-to-use computer**.

Similarly, a data appliance is a **ready-to-use optimized data system**.

Big Data Security

Protecting Big Data against:

- Unauthorized access
- Modification
- Theft
- Leakage
- Attacks

example

Bank stores:

Customer name
Account number
Transaction history
Credit card information

An attacker tries to access it.

Security mechanisms prevent unauthorized access.

CIA Triad

Confidentiality

Who can see the data?

Example:

Only authorized bank employees can access customer records.

Integrity

Has the data been changed incorrectly?

Example:

Bank balance:

50,000 → 5,000

If an attacker changes it, integrity is violated.

Availability

Can authorized users access the data when needed?

Example:

Bank server crashes and customers cannot access their accounts.

→ Availability problem.

Authentication vs Authorization

Authentication

WHO are you?

Example:

Login using:

Username + Password

Authorization

WHAT are you allowed to do?

Example:

Student → View marks
Teacher → Enter marks
Admin → Modify records

Easy memory

Authentication = Who?

Authorization = What can you do?

Big Data Issues

Storage

Where do we store huge amounts of data?

Processing

How do we process huge datasets quickly?

Scalability

What happens when data grows?

Security

How do we protect it?

Privacy

How do we protect personal information?

Data Quality

Is the data accurate?

Integration

How do we combine data from different sources?

Real-time processing

How do we process continuously arriving data quickly?

Big Data Applications

Healthcare

Patient Data
MRI
ECG
Medical Records
IoT Sensors
↓
Big Data Analytics
↓
Disease prediction

Banking

Transactions
↓
Big Data Analytics

↓
Fraud Detection

Amazon/Retail

Customer behavior
↓
Analytics
↓
Product Recommendation

Transportation

GPS + Traffic Sensors
↓
Analytics
↓
Traffic Prediction

Social Media

Posts + Likes + Comments
↓
Analytics
↓
Trend/Sentiment Analysis

